Justin Do

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EDUCATION AND CLINICAL PERFORMANCE

Doctor of Medicine Jul 2023 - Present

Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, PA

- USMLE Step 2 CK: 261; USMLE Step 1: Pass
- Class rank: Top third of class
- Core clerkship honors (8/9): Anesthesiology, IM, Neurology, EM, OB/GYN, Pediatrics, Family Medicine, Psychiatry
- Fourth-year honors: UCLA Anesthesiology Sub-Internship; Jefferson Regional Anesthesia

Bachelor of Science, Premedicine, cum laude Aug 2020 - May 2022

The Pennsylvania State University, State College, PA

- Accelerated 7-year BS/MD pathway; GPA: 3.90; Dean's List

HONORS AND AWARDS

Selected Honors

- Hobart Amory Hare Honor Society, 2026
- SKMC Dean's Research Award, 2025
- James B. Erdmann Student Team Award for Excellence in Interprofessional Education and Collaborative Practice, 2025
- Arthur Johnson, MD Visiting Clerkship Scholarship, Cedars-Sinai Medical Center, 2026
- 2nd Place (tie), Medical Student Best Poster Presentation, 2024 FMEC Annual Meeting
- 2nd Place, Student Investigator Award (Oral Presentation), Alpha Omega Alpha Research Symposium, SKMC, 2025
- 1st Place, Outstanding Oral Presentation (Design Track), Scholarly Inquiry Research Project Symposium, SKMC, 2025

RESEARCH EXPERIENCE

Research Fellow Sep 2024 - Present

Department of Anesthesiology and Perioperative Medicine, Thomas Jefferson University

- Initiated and led perioperative machine-learning and clinical-informatics projects with clinical and computer-science collaborators from study design through publication
- Led a 57,055-operation study of interpretable machine learning for postoperative AKI, resulting in a first-author *JAMIA Open* article and ANESTHESIOLOGY 2025 oral presentation
- Led a 122,481-operation MACCE prediction study by adapting the AKI pipeline; ANESTHESIOLOGY 2026 ePoster
- Led a 3,460-case study of intraoperative waveforms for postoperative AKI and ICU risk; ANESTHESIOLOGY 2026 ePoster
- Led a 13,086-image benchmark of conversational segmentation across seven medical modalities, including a skin-tone fairness audit; medRxiv preprint and IARS/SOCCA 2026 poster
- Initiated and led a study of 4,131 FDA anesthetic-device authorizations; coordinated classification and expert adjudication, led analysis and manuscript drafting, and presented an IARS/SOCCA 2026 poster
- Linked and curated Epic and ROTEM data with liver-transplant clinicians for a perioperative informatics study
- Served as a peer reviewer for *JAMIA Open* (2026)

Research Intern Aug 2023 - Present

Health Design Lab, Thomas Jefferson University

- Led interdisciplinary health-design projects in multimodal AI, anatomical modeling, and rehabilitation prototyping
- Led a multimodal LLM project that analyzed home photographs for fall hazards, generated CDC STEADI-grounded recommendations, and visualized proposed changes
- Proposed and led the shift from inpatient falls to home prevention; coordinated clinical input and 54 paired evaluations, then led analysis, figures, and drafting for a first-author *JMIR Aging* paper
- Created CT-based anatomical models for an interventional-radiology training prototype
- Advised rehabilitation projects on CAD and FDM 3D-printing feasibility

Interdisciplinary Clinical Research Contributor 2024 - Present

Departments of Neurosurgery, Vascular Surgery, Family Medicine, and Oncology, Thomas Jefferson University

- Co-authored five published papers across neurosurgery, vascular surgery, family medicine, and oncology, contributing to study design, analysis, technical implementation, and manuscript development
- Wrote CDC WONDER analysis scripts and produced all tables and figures for a national study of place-of-death disparities among people with cerebral atherosclerosis, published in the *Journal of Clinical Neuroscience*
- Recoded chart-review data and analyzed associations among management approach, community distress, and follow-up after thoracoabdominal aortic aneurysm; revised abstracts and the *American Journal of Medical Quality* manuscript

Interdisciplinary Clinical Research Contributor (continued) 2024 - Present

- Helped develop the visual-analysis framework and independently coded medical-student cartoons examining the transition to clerkships; contributed to theme development and drafting the *Medical Science Educator* article
- Reframed a vascular patient-education study around source-grounded LLM rewriting; implemented the technical workflow and contributed to drafting and revising the *JVS-Vascular Insights* article
- Conducted qualitative coding and drafted major sections of the initial manuscript for an *AI in Precision Oncology* study of patients' views of AI-assisted portal communication

Research Assistant Aug 2021 - Jun 2024

ReMind Lab, Penn State College of Nursing

- Co-authored four publications on aging expectations, subjective cognition, patient-provider communication, and modifiable dementia risk through literature synthesis, manuscript drafting, and revision
- Drafted introduction and discussion sections for a *BMC Geriatrics* study of patient-provider conversations about cognition
- Led the literature update and initial draft for a first-author *Journal of Gerontological Nursing* editorial translating modifiable dementia-risk evidence into conversations with older adults
- Reviewed the literature, interpreted results, prepared manuscript tables, and drafted sections for an *Aging & Mental Health* study of aging expectations and subjective cognition
- Designed search terms; completed screening, extraction, verification, CASP appraisal, and synthesis tables; and drafted sections and reviewer responses for a *Gerontology* systematic review of views of aging and subjective cognition

Clinical Research Coordinator Associate May 2022 - Oct 2022

Patel Lab, Stanford University

- Supported startup of NCT05297734, a 24-site trial comparing technology- and team-based delivery of supportive cancer care
- Configured and stress-tested REDCap, updated ClinicalTrials.gov, supported initial site launches including a VA, and enrolled and interviewed participants under protocol
- Created and presented the first-author protocol poster at the 2022 Stanford Oncology Annual Scientific Conference

TEACHING, MENTORSHIP, AND SERVICE

FIRST Tech Challenge Mentor / Game Manual 0 Managing Editor Jul 2020 - Present

Game Manual 0; Team 9656; AllStars Education Foundation

- Co-authored Game Manual 0, writing approximately 80-100 pages on drivetrains, build systems, and mechanical design
- Served as one of six managing editors; the guide reached five language editions and approximately 400,000 annual visits
- Mentored Team 9656 through design reviews, troubleshooting, control-system debugging, and advising student leaders
- Helped design a middle-school robotics pathway, goBILDA mini-bot curriculum, parts lists, and equipment plan

Planetary Health Report Card Student Contributor Jan 2024 - Apr 2026

Planetary Health Report Card, Sidney Kimmel Medical College

- Co-authored three SKMC Planetary Health Report Cards (2024-2026) and two related SHARE posters (2024-2025)
- Researched assigned sections on buildings, food, transportation, recycling, and housing across three annual cycles
- Contacted institutional personnel to gather information and drafted scored narratives with supporting citations

Student Publications Editor and Writer Feb 2024 - Mar 2025

Inside Out (student arts and humanities journal); Beyond the Airway (student-run anesthesiology journal), Thomas Jefferson University

- Reviewed and discussed poems, stories, paintings, and photography for Jefferson's student arts and humanities journal
- Assessed anesthesiology competitiveness through literature, national reports, and match data in a student journal article

Science Outreach Educator and Volunteer Aug 2023 - Sep 2024

JeffTeach; Anato-Bee Eastern Regional Competition, Thomas Jefferson University

- Adapted and delivered hands-on elementary-science lessons through a medical student-led outreach program
- Served as a standardized patient, response grader, campus guide, and mentor at a high-school anatomy competition

Student Interviewer Aug 2023 - Jan 2025

Sidney Kimmel Medical College, Thomas Jefferson University

- Interviewed SKMC and PMM BS/MD applicants using separate rubrics; documented current-student observations

Treasurer Jan 2024 - Jan 2025

ARTICLE Research Club, Thomas Jefferson University

- Administered the annual budget, coordinated events, and helped students identify research opportunities and mentors

PROFESSIONAL MEMBERSHIPS AND SKILLS

Professional societies: ASA Student Member; IARS Student Member.

Technical skills: R, Python/PyTorch, REDCap, Covidence, Zotero, Materialise Mimics segmentation, NSF ACCESS cloud computing, Fusion 360 rendering/visualization, FDM 3D printing.

Interests: science-fiction worldbuilding, aerospace history, PC building/game modding, character-driven shows, StairMaster workouts, and neighborhood restaurants with friends.

SCHOLARLY WORK - Published / Accepted / In Press

1. **Do J**, Shah KH, Xu MC, Kim AH, Suresh V, Guggilla N, Li M, Kothari R. Integration of intraoperative data in interpretable machine learning models to predict postoperative AKI in noncardiac surgery patients. JAMIA Open. 2026 Jun 20;9(3):ooag092. doi: 10.1093/jamiaopen/ooag092. PMID: 42327625; PMCID: PMC13282712
2. **Do J**, Suresh V, Zhang L, Chavre BM, Cha J, Pugliese R. Leveraging multimodal large language models for fall risk reduction in older adults in the home: proposed model design. JMIR Aging. 2026;9:e77591. doi:10.2196/77591. PMID:42127229. PMCID: PMC13170740
3. Xu MC, Schwartz D, **Do J**, Abai B, Salvatore D, DiMuzio P, Nooromid M. Management modality of thoracoabdominal aortic aneurysms is associated with lower rates of follow-up. Am J Med Qual. 2026;41(4):221-222. doi:10.1097/JMQ.0000000000000307. PMID:42358180.
4. Xu MC, Habash N, **Do J**, et al. Large language models improve readability of patient education materials on vascular conditions. JVS Vasc Insights. 2026;4:100318. doi:10.1016/j.jvsvi.2025.100318
5. Fuleihan AA, Roy JM, **Do J**, et al. Understanding disparities in cerebral atherosclerosis mortality across the United States: an analysis of the CDC WONDER database. J Clin Neurosci. 2025;140:111532. doi:10.1016/j.jocn.2025.111532. PMID:40753666
6. Fishman C, **Do J**, Markham F. Drawing it out: a descriptive study on how medical students use graphic medicine to depict their transition to third year. Med Sci Educ. 2025;35(4):1855-1859. doi:10.1007/s40670-025-02377-w. PMID:41112889
7. Hazlewood C, **Do J**, Logan JA, et al. Patient perceptions of electronic patient portal artificial intelligence use at an urban medical center. AI Precis Oncol. 2025;2(4):142-148. doi:10.1177/2993091X251364897
8. Hill NL, **Do J**, Bratlee-Whitaker E, et al. Views of aging and subjective cognition in middle-aged and older adults: a systematic review. Gerontology. 2025;71(1):49-70. doi:10.1159/000542507. PMID:39586249
9. Hill NL, Bhargava S, **Do J**, et al. Just as expected? Older adults' aging expectations are associated with subjective cognition. Aging Ment Health. 2025;29(3):444-451. doi:10.1080/13607863.2024.2399080. PMID:39241125
10. **Do J**, Hill NL. Reducing dementia risk: the latest evidence to guide conversations with older adults. J Gerontol Nurs. 2023;49(9):3-5. doi:10.3928/00989134-20230815-01. PMID:37650850
11. Hill NL, Bratlee-Whitaker E, Jang H, Bhargava S, Sillner AY, **Do J**, Mogle J. Patient-provider communication about cognition and the role of memory concerns: a descriptive study. BMC Geriatr. 2023;23(1):342. doi:10.1186/s12877-023-04053-3. PMID:37259029

SCHOLARLY WORK - Published Abstracts

1. **Do J**, Guggilla N, Suresh V, Kothari R. Evaluating conversational image segmentation for medicine: performance, failure modes, and a fairness audit across seven modalities [abstract]. Anesth Analg. 2026;142(5S):1134. Supplement doi:10.1213/01.ane.0001282976.51283.05.
2. **Do J**, Fuleihan A, Wenyon S, Menta A, Xu M, Joseph J. Anesthetic device innovation from 2000 to 2023: a comprehensive analysis of FDA authorization trends from 4,131 approvals by device category, regulatory pathway, and approval time [abstract]. Anesth Analg. 2026;142(5S):312. Supplement doi:10.1213/01.ane.0001282976.51283.05.

SCHOLARLY WORK - Submitted / Under Review / Preprints

Do J, Guggilla N, Suresh V, Kothari R. Evaluating conversational image segmentation for medicine: performance, failure modes, and a fairness audit across seven modalities. medRxiv. Preprint posted online November 23, 2025:2025.11.21.25340008. doi:10.1101/2025.11.21.25340008

SCHOLARLY WORK - Presentations

1. **Do J**. More Data, More Information? What AI Actually Learns from the OR. Invited Power Talk accepted for presentation at: Society for Technology in Anesthesia Annual Meeting; January 17, 2027; Scottsdale, AZ.
2. **Do J**, Guggilla N, Kim AH, Suresh V, Li M, Kothari R. Integration of intraoperative data for the prediction of MACCE using deep learning. ePoster presented at: ANESTHESIOLOGY 2026; October 17, 2026; San Diego, CA.
3. **Do J**, Guggilla N, Wingert TE, Kothari R. Interpretable machine learning of intraoperative waveforms for AKI and ICU risk discrimination. ePoster presented at: ANESTHESIOLOGY 2026; October 17, 2026; San Diego, CA.
4. **Do J**, Guggilla N, Suresh V, Kothari R. Evaluating conversational image segmentation for medicine: performance, failure modes, and a fairness audit across seven modalities. Poster presented at: IARS/SOCCA 2026 Annual Meeting; May 2, 2026; Montreal, QC, Canada.
5. **Do J**, Fuleihan A, Wenyon S, Menta A, Xu M, Joseph J. Anesthetic device innovation from 2000 to 2023: a comprehensive analysis of FDA authorization trends from 4,131 approvals by device category, regulatory pathway, and approval time. Poster presented at: IARS/SOCCA 2026 Annual Meeting; May 3, 2026; Montreal, QC, Canada.
6. **Do J**, Kim A, Suresh V, Guggilla N, Shah KH, Xu MC, Li M, Kothari R. Using intraoperative data via machine learning to predict postoperative AKI in noncardiac surgery. Oral and ePoster presentation presented at: ANESTHESIOLOGY 2025; October 13, 2025; San Antonio, TX.
7. Xu MC, Habash N, **Do J**, Salvatore D, DiMuzio P, Nooromid M, Abai B. Large language models improve readability of patient education materials on vascular conditions. Poster presentation presented at: Eastern Vascular Society 39th Annual Meeting; September 4-7, 2025; Nashville, TN.
8. Driben M, Guler M, MacMillan R, Flomenbaum S, Matheson M, Gormus M, Gelfer S, Atrafi T, **Do J**, Liu J, Bartz J, Dale S. 2024-2025 Planetary Health Report Card for Sidney Kimmel Medical College at Thomas Jefferson University. Poster presentation presented at: SHARE 2025; June 20, 2025; Online.
9. Xu MC, Schwartz D, **Do J**, Zack J, Abai B, Salvatore D, DiMuzio P, Nooromid M. Distressed Community Index scores are not associated with differential TAA presentation, follow-up, and management. Poster presentation presented at: Society for Clinical Vascular Surgery; March 29-April 2, 2025; Austin, TX.

10. **Do J**, Zhang L, Suresh V, Cha J, Chavre B, Pugliese R. Stability Sidekicks: leveraging multimodal AI for automated fall risk assessment in elderly patients' homes. Oral presentation presented at: Jefferson AOA Research Symposium; February 12, 2025; Philadelphia, PA.
11. Xu MC, **Do J**, Yang H, Salvatore D, Abai B, DiMuzio P, Nooromid M. The effect of preoperative anemia on postoperative length of stay in open replacement of abdominal aorta: insights from the INSPIRE database. Oral presentation presented at: Jefferson AOA Research Symposium; February 12, 2025; Philadelphia, PA.
12. Chavre B, Cha J, **Do J**, Zhang L, Pugliese R. Reoptimizing fall prevention kits to increase fall risk awareness in older adults. Poster presentation presented at: Jefferson AOA Research Symposium; February 12, 2025; Philadelphia, PA.
13. Schwartz D, Xu MC, **Do J**, Zack J, Abai B, DiMuzio P, Nooromid M. Nonsurgical management of TAAAs is associated with greater rates of being lost to follow-up. Oral and poster presentation presented at: 4th Annual Philadelphia Cardiovascular Congress; October 24, 2024; Philadelphia, PA.
14. Fishman C, **Do J**, Markham F. When mentors become monsters: using cartoons to explore hierarchy and feelings of incompetence in medical training. Poster presentation presented at: Family Medicine Education Consortium Annual Meeting; September 21, 2024; Philadelphia, PA.
15. Guler M, Ansel D, Mehta N, **Do J**, Atrafi T, Smith E, Owrey M, Han J, Owen-Oliner L, Hostetler J, Archer E, Driben M, Vanichkachom AC, Wahba N, Deshpande A, DeSimone K, Dale S. Planetary Health Report Card for Sidney Kimmel Medical College at Thomas Jefferson University. Poster presentation presented at: SHARE 2024; June 25, 2024; Online.
16. Fishman C, **Do J**, Bashyal R, Hakorimana F. Knowledge and receptiveness of epidural labor analgesia by parturients at two teaching hospitals in Rwanda: a cross-sectional study. Poster presentation presented at: Health Equity and Quality Improvement (HEQI) Summit; May 3, 2024; Philadelphia, PA.
17. **Do J**, Fishman C, Bratlee-Whitaker EY, Jang H, Bhargava S, Sillner AY, Mogle J, Hill NL. Patient-provider communication about cognition and the role of memory concerns: a descriptive study. Poster presentation presented at: Equity in Medical Education Summit; April 2, 2024; Philadelphia, PA.
18. **Do J**, Fishman C, Bhargava S, Bratlee-Whitaker EY, Turner JR, Mogle J, Hill NL. Aging expectations predict subjective cognitive decline among community-dwelling older adults. Poster presentation presented at: Jefferson AOA Research Symposium; February 9, 2024; Philadelphia, PA.
19. **Do J**, Medrano HS, Wood EH, Mathew MS, Agrawal M, Kapphahn KI, Patel MI. Protocol for a comparative effectiveness trial of two supportive cancer care delivery models for adults with cancer at 24 sites nationwide. Poster presentation presented at: Stanford Oncology Annual Scientific Conference; October 19-22, 2022; Pacific Grove, CA.
20. **Do J**, Seitam P, Noga R, Turner JR, Bratlee-Whitaker EY, Hill NL. Exploring associations among age category, personality, and social desirability in older adults. Poster presentation presented at: The Pennsylvania State University Annual Undergraduate Research Exhibition; April 8, 2022; University Park, PA.
21. Bratlee-Whitaker EY, Hill NL, Mogle J, **Do J**. Personality moderates affective responses to daily memory lapses in middle-aged and older adults without cognitive impairment. Poster presentation presented at: Virtual Scientific Meeting of the Society for Affective Science; March 31, 2022; Virtual.

SCHOLARLY WORK - Other Educational Products

1. Gelfer S, Bartz J, Driben M, Manca E, Whitcomb M, Brown H, Atrafi T, **Do J**, Zarrinnegar D, Deshpande A, Uppal M, Dale S. Planetary Health Report Card for Sidney Kimmel Medical College at Thomas Jefferson University. Planetary Health Report Card 2026. Published April 2026.
2. Driben M, Guler M, MacMillan R, Flomenbaum S, Matheson M, Gormus M, Gelfer S, Atrafi T, **Do J**, Liu J, Bartz J, Dale S. Planetary Health Report Card for Sidney Kimmel Medical College at Thomas Jefferson University. Planetary Health Report Card 2025. Published April 2025.
3. **Do J**. The National Resident Matching Program (NRMP) Match: Is Anesthesiology Becoming More Competitive? Beyond the Airway. 2024;1(1):11-12. https://jdc.jefferson.edu/beyond_the_airway/vol1/iss1/1
4. Guler M, Ansel D, Smith E, Han J, Owrey M, Mehta N, Owen-Oliner L, Hostetler J, Archer E, Driben M, **Do J**, Atrafi T, Vanichkachom AC, Wahba N, Deshpande A, DeSimone K, Dale S. Planetary Health Report Card for Sidney Kimmel Medical College at Thomas Jefferson University. Planetary Health Report Card 2024. Published April 2024.
5. Game Manual 0 Contributors. Game Manual 0: A Comprehensive Guide to FIRST Tech Challenge Robotics. Published online; continuously updated. https://gm0.org. Role: **Do J**, managing editor and contributor.